

Sprint 3 Submission - PetAlert (Team Sunrise Inc)

GitHub Repository: <https://github.com/SunriseInc-Capstone/PetAlert>

Team Information and Introduction

Team Name

Sunrise Inc

Project Name

PetAlert

Team Members

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Overview

Problem Context

Pet owners face confusion and lost time when a pet goes missing. Information such as photos, contacts, and identification details is often scattered across different apps or papers, making response slower and less organized.

Goal

PetAlert provides a **local-first mobile app** that stores pet profiles, emergency contacts, and alert templates in one place. With a single tap, users can generate and share a missing-pet alert via social media, email, SMS, or printable flyer—helping reunite pets and families faster.

Requirements

Functional Requirements (User Stories)

- R1 Pet Profile: Create/update pet profiles with photo and details (local storage).

- R2 Emergency Contacts: Add/edit vet, shelter, police contacts with one-tap call/email/SMS.
- R3 Missing Pet Alert: Generate social media post, email/SMS draft, and phone script instantly.
- R4 Regular Update Prompts: Local notifications reminding owners to update info/photos.
- R5 Photo Gallery: Upload/view multiple pet photos locally.
- R6 Recommendations: Static safety tips and search strategies.
- R7 Toxic Foods Infographics: Quick visual guides for dangerous foods.

Non-Functional Requirements

- NF1 Security: All data stored locally; optional OS-level lock (FaceID/Passcode).
- NF2 Reliability: Core features work offline (profiles, contacts, alerts).
- NF3 Usability: ≤3 taps to reach any function; emergency-first button on home.
- NF4 Accessibility: Support large text, high contrast, screen reader labels.
- NF5 Cross-Platform: Runs consistently on iOS 14+ and Android 10+.
- NF6 Accuracy: Generated alerts match stored details exactly.

Design

The app uses a tab-based navigation system:

Home: Prominent “Missing Pet Alert” button.

Pets: Displays all pet profiles with add/edit options.

Pet Detail: Shows stored information and allows generating an alert.

Contacts: Emergency contacts with instant action buttons.

Toxic Foods: Scrollable infographic cards and quick-reference visuals.

Reflection Report

Accomplishments

Sprint 3 marked a major transition from a fully offline prototype to a hybrid local + cloud-enabled application. The team successfully implemented authentication, strengthened app structure, and fixed several architectural limitations from earlier sprints.

Key accomplishments include:

- Integrated Firebase Authentication (email/password signup + login) and added navigation redirection for authenticated users.
- Fully implemented Missing Pet Alert module with create, view, and edit screens, including local storage via JSON and SharedPreferences.
- Improved UI design across Home, Pet Profiles, Add Pet, and Missing Alerts using consistent theming, icons, and Material 3 components.
- Resolved multiple build issues on Android and macOS (FlutterFire setup, Gradle errors, Google services configuration).
- Established clear file structure with separation of concerns: models/, services/, auth/, and feature-specific folders.
- Pushed all Sprint 3 code to GitHub with proper version history and documented changes.

Overall, Sprint 3 significantly increased the application's professionalism, scalability, and reliability.

What Went Well

Team collaboration continued to be strong, especially during the technical setup of Firebase and FlutterFire.

- Members communicated consistently while debugging Android/Mac issues.
- The UI/UX workflow improved after reviewing real device/emulator behavior.
- The team adapted quickly to new concepts (Firebase, async logic, JSON storage).
- Everyone contributed meaningfully to feature testing and documentation.

As a result, the sprint maintained productive momentum despite the increased complexity.

What Could Be Improved

Despite major improvements, some challenges remain:

- **Firebase setup consumed more time than expected**, causing delays in feature implementation.
- UI testing on both Android and iOS should begin earlier to avoid last-minute debugging.

Plans for Sprint 3.5

During this sprint, we will resume the work initiated in Sprint 3 and continue developing our planned features (emergency contacts, pet alert, photo gallery, toxic food), and if necessary, do more research.

Rubric Reflection

What went well in the presentation?

The Sprint 3 presentation showcased a major leap in functionality.

- Demonstrated the successful **login and signup flow** using Firebase.
- Showed the improved UI design and smoother navigation experience.
- Displayed completed features such as Pet Profiles, Add Pet with image picker, and Missing Alerts.

Each team member explained their part clearly, and the prototype looked much closer to a production-ready app.

What can be improved in the presentation?

What we can improve in this presentation is to come prepared with a backup demo in the form of pictures or a video. We also faced an internet problem at the beginning of the presentation, which was later solved. We can use a phone hotspot next time to avoid a delay in the demo with the professor.

Other Notes

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